Tidal potential measured with distant optical clocks

Lei Hou[1,2], Xiang Zhang[1,2,3], Qiang Zhu[4], Xiang-Pei Liu[1,3], Fa-Xi Chen[1,5], Wen-Xuan Song[1,2,3], Zhuan-Xian Xiong[1,4], Jian-Yu Guan[1,2], Yu-Chen Zhang[1,2,3], De-Quan Kong[1,2,3], Jia-Hao Zhang[1,2,3], Long-Fei Liao[1,2,3], Shao-Mao Wang[1,2], Bing Wang[4], De-Zhi Xiong[1,4], Xing-Yang Cui[1,2], Ping Xu[1,2,3], Peng Liu[1,2], Jie Li[2,3], Xiao Jiang[1,2,3], Hua-Qing Zhang[1,5], Ling-Xiang He[1,4], Li-Bo Li[5], Qi Shen[1,2,3], Youchao Xie[6], Lei Wang[6], Sheng-Kai Liao[1,2,3], Juan Yin[1,2,3], Cheng-Zhi Peng[1,2,3], Wen-Bin Shen[6], Baolong Lyu[1,4], Han-Ning Dai[1,2,3], Qiang Zhang[1,2,3,5], Yu-Ao Chen[1,2,3], Hai-Feng Jiang[1,2,3], Jian-Wei Pan[1,2,3]

[1] Hefei National Laboratory, University of Science and Technology of China, Hefei, China

[2] Shanghai Research Center for Quantum Sciences and CAS Center for Excellence in Quantum Information and Quantum Physics, University of Science and Technology of China, Shanghai, China

[3] Hefei National Research Center for Physical Sciences at the Microscale and School of Physical Sciences, University of Science and Technology of China, Hefei, China

[4] Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences, Wuhan, China

[5] Jinan Institute of Quantum Technology and CAS Center for Excellence in Quantum Information and Quantum Physics, University of Science and Technology of China, Jinan, China

[6] Time and Frequency Geodesy Center (TFGC), School of Geodesy and Geomatics, Wuhan University, Wuhan, China

The Moon and the Sun alter Earth's gravitational potential through their tidal fields and the responses of the solid Earth and oceans. Changes in the tidal potential modify the relative ticking rates of clocks at different locations. Advances in optical clocks and fiber-optic network technologies have made clocks precise probes for probing spacetime, providing an experimental foundation for the long-term vision of monitoring Earth's gravitational potential with a global clock network. In this work, we use strontium and ytterbium clocks separated by 692 km to detect tidal variations in the potential difference between the coastal region of Shanghai and the inland region of Wuhan. Based on 545.9 hours of observations spanning 353 days, the joint measurement of solid-Earth and ocean loading agrees with model predictions with a consistency of 0.96 ± 0.11. Using spectral analysis, we separated the principal lunar semidiurnal tidal constituent M_2 and the principal solar semidiurnal tidal constituent S_2, with frequencies of 22.361 μHz and 23.147 μHz, respectively, and signal-to-noise ratios of 16.7 dB and 9.3 dB, respectively. This experiment demonstrates that optical clock networks can already be used for high-precision gravitational potential calibration and detection, laying a foundation for related applications.

Introduction

Measurements of Earth's gravity field reveal how water, air, and the solid Earth redistribute mass. Ground-based gravimetry and gradiometry measure gravitational acceleration and its spatial gradients, respectively, while precise leveling combined with gravity observations determines potential differences.[1,2] Satellite missions have tracked large-scale changes in water storage and the climate system.[3,4] Monitoring time-varying

potential differences between distant stations requires resolving weak signals and distinguishing contributions with similar temporal characteristics.[5] Tides provide a well-defined test case for this challenge. The tidal fields of the Moon and Sun act together with solid Earth deformation and the redistribution of ocean water. Atmospheric pressure tides also drive the oceans.[6] The solid Earth tidal potential can reach amplitudes of several $\mathrm{m}^2/\mathrm{s}^2$ at individual stations, and resolving smaller contributions requires higher measurement sensitivity.[7]

Optical clocks provide another means of measuring potential differences through their relative clock rates. Near the Earth's surface, a relative frequency change of 10^{-18} corresponds to an equivalent height of about 1 cm.[8] Experiments have resolved gravitational redshifts on laboratory scales and measured static potential differences between distant sites.[2,9–11] The fractional systematic uncertainty of optical clocks is now approaching 10^{-19}, and independent lattice-clock experiments have achieved instabilities of a few $10^{-17}/\sqrt{\tau/\mathrm{s}}$, where τ is the averaging time.[12,13] Non-tidal mass changes in the atmosphere, oceans, and continental waters may be significant at the 10^{-18} fractional frequency level.[7] Advances in fiber-optic networks and free-space frequency transfer are extending clock comparisons to long distances.[14–18] These advances are driving the development of optical-clock networks for monitoring time-varying gravitational potentials.[5,8] However, resolving neighbouring tidal components requires sufficient sensitivity and temporal coverage, especially in the presence of observational gaps.

Here we detect tidal variations in the potential difference between a strontium clock in coastal Shanghai and a ytterbium clock in inland Wuhan. The clocks are separated by 692 km and connected by a 1,350-km fiber-optic link.[19] Using 545.9 hours of observations distributed over 353 days, we combine variations within observing segments while allowing each segment an independent frequency offset. We jointly estimate the responses to two neighbouring semidiurnal tidal groups, M_2+N_2 and S_2+K_2, with phases and relative amplitudes fixed within each group.

Results

Tidal potential on a coastal–inland baseline

The Shanghai Sr and Wuhan Yb clocks measure a tidal potential difference over a 692-km baseline, linked by 1,350 km of phase-stabilized fibre (Fig. 1a). Five campaigns between September 2025 and August 2026 provide approximately 544 h of retained observations in 54 segments. The measured potential combines the solid-Earth tide and ocean loading. At M_2, their predicted differential amplitudes are 3.41×10^{-18} and 1.42×10^{-18} in fractional frequency. Their 20° phase difference gives a combined amplitude of 4.77×10^{-18} (Fig. 1b). The calculation follows the IERS conventions[20] and uses FES2022[21,22] with a loading Green's function that includes water attraction, induced potential and clock displacement.[23] The redistribution of seawater affects the two stations differently. Both its gravitational attraction and the Earth's response enter the potential sampled by each clock; treating loading as a change in clock height alone would omit part of the measured quantity. The coastal–inland geometry makes the differential

loading contribution appreciable, while the two components add with the phase relation shown in Fig. 1b.

The joint fit gives response factors of 0.97 ± 0.14 for the lunar group (M_2, N_2) and 0.90 ± 0.29 for the near-12-h group (S_2, K_2), consistent with the predicted combined tide (Fig. 2; Extended Data Fig. 11). The groups retain their astronomical phases and internal amplitude ratios. Allowing the lunar phase to vary gives an amplitude of 0.98 ± 0.11 and a phase offset of $-6^\circ \pm 8^\circ$, supporting the predicted lunar timing. Data and predictions undergo the same 1,200-s triangular filter at 600-s steps, with one offset per segment. The quoted uncertainties are statistical standard deviations, with their calibration specified in Methods. A response of one denotes agreement with the predicted amplitude. Segment offsets absorb constant differences between acquisition intervals, so the estimates are determined by changes within the retained records. Applying the filter to both the measured ratio and the predicted signal accounts for the averaging when estimating the response. The combined template gives a response of 0.90 ± 0.12, while the separate groups establish how that agreement is distributed between neighbouring tidal contributions.

The summer campaign supplies the lunar–solar phase coverage needed to distinguish the neighbouring responses (Fig. 1d). Its 274 h span two months and the full S_2–M_2 relative-phase cycle, providing 61% of the conditional information on the lunar estimate. The summer lunar response is 0.87 ± 0.18, compared with 1.05 ± 0.31 for the four earlier campaigns together. Campaign estimates agree ($\chi^2 = 4.1$ for four degrees of freedom, $p = 0.40$). Figure 2 shows the summer response; Extended Data Fig. 4 retains the calendar dates and gaps of the April record. During a short campaign, the lunar oscillation occupies a limited range of phases relative to the solar day. Extending the observations across that relative-phase cycle changes the relation between the lunar and solar-frequency waveforms, allowing their contributions to be estimated with less ambiguity. The two subsets therefore give compatible lunar responses despite their different phase coverage. The summer phase-folded observations also follow the predicted oscillation after removal of the fitted non-lunar terms. Their error bars describe the conditional uncertainty of the displayed means; the response uncertainties above come from the joint estimates.

Spectral density and separate detection

The equivalent-height power spectral density resolves contributions at the M_2 and S_2 frequencies, 22.36428 and 23.14815 μHz (Fig. 3). At these frequencies, the conditional densities are 1.28×10^{-3} and 1.94×10^{-4} $\mathrm{m}^2/\mu\mathrm{Hz}$, respectively, with the other principal component, segment offsets and freely phased N_2 and K_2 terms controlled. Both values use the same 3,299 non-overlapping 600-s means, representing 545.924 member-hours, with equal weights and the normalization defined in Methods. Equivalent height expresses the measured potential variation as a length through $h_{\mathrm{eq}} = c^2 y / g_0$, where y is fractional frequency and g_0 is standard gravity. The spectra therefore place the two contributions on the same gravitational scale. The non-overlapping reduction retains the observation gaps and uses the stored bin-centre times; its sample selection is

distinguished from the triangular-filter reduction in Methods. Conditional scans retain each contribution when the other principal frequency is controlled (Fig. 3d). At every trial frequency, the competing component has free cosine and sine coefficients, so its amplitude and phase can adjust. The target response is then estimated from the variation left after that adjustment. The reported densities are evaluated at the astronomical frequencies, using symmetric controls for the two components. Subtracting the jointly fitted M_2 waveform leaves an S_2 sinusoidal amplitude of 1.921×10^{-18}, compared with 2.918×10^{-18} before subtraction. The reciprocal subtraction leaves M_2 at 4.505×10^{-18}, compared with 4.857×10^{-18} (Fig. 3b,c; Extended Data Fig. 8). The projected sine–cosine subspaces have a maximum canonical correlation of 0.290; the maximum absolute cross-component coefficient correlation is 0.261 under UTC-day deletion (Extended Data Fig. 9). The subtracted waveforms come from the joint fit and are held fixed while the remaining target is re-estimated. Subtracting both gives the residual spectrum in Fig. 3a. The correlations quantify how the sampled waveforms overlap and how the estimated coefficients vary together. Together with the grouped responses, these results support separate detection of the two frequency contributions and quantify their coupling. The free-phase S_2 amplitude is less precise than M_2; their joint coefficient regions and amplitude ranges are reported in Methods and Extended Data Fig. 12. Here solar-frequency identifies the 12-h component; its frequency alone does not distinguish an astronomical S_2 tide from a laboratory variation at the same period.

The sampling response explains specific neighbouring peaks. Single-component injections generate M_2 sidelobes at 22.1016 and 22.6283 µHz. Subtracting the fitted M_2 waveform reduces the corresponding observed spectral densities by 99.56% and 99.39%, with minima near unit compensation (Extended Data Fig. 10). The matching locations and subtraction responses identify M_2 leakage at these frequencies. Other neighbouring features have different compensation responses, as shown by the individual sampling calculations. Each injection consists of one fitted sinusoid evaluated only at the measured times. Passing it through the same offset and neighbouring-frequency treatment shows where the observation window places power away from the injected frequency. Varying the subtracted waveform's amplitude then tests the corresponding feature in the measured spectrum. The observed compensation minima agree with the amplitudes expected from the fitted M_2 waveform. This comparison explains why several visible peaks can accompany one physical oscillation and why peak positions alone are insufficient to count components.

Measurement reliability

The lunar response remains consistent when solar-day terms are included (Fig. 4). Common 24-h and 12-h terms give 1.00 ± 0.13; campaign-specific 12-h terms give 0.82 ± 0.17, or 0.86 ± 0.18 with 24-h terms also included. A 10^{-18} sinusoidal perturbation at either period changes the lunar response by at most 0.020 if its phase is fixed across the record. Independent campaign phases increase the maxima to 0.032 at 24 h and 0.12 at 12 h. A common-phase disturbance repeats with the same phase in all campaigns, whereas the campaign-specific case allows its timing to change between campaigns. Among these

specified disturbances, the campaign-dependent 12-h term produces the largest change in the lunar estimate. The campaign-specific fit allows those daily contributions to vary in the measured record and still gives a lunar response consistent with the main fit. The perturbation calculation and the fitted daily terms thus describe both the estimator's susceptibility and the response obtained when those terms are included.

Environmental records, link monitoring and alternative processing support the stability of the measurement. On the reported 474-h overlap with the fibre round-trip monitor, including the monitor changes the lunar and whole-template coefficients by less than 0.005 on matched clock samples. Temperature regressions provide a further check on residual blackbody corrections. Both clocks apply blackbody corrections during operation. The Sr chamber record averages readings from 20 temperature sensors and overlaps 98 h of the filtered summer data. Regressing the clock ratio on the calculated blackbody shift gives a coefficient of 0.92 ± 0.74, or 0.72 ± 0.74 after excluding the chamber-temperature transient, consistent with zero at this precision. These regressions test remaining errors proportional to the measured chamber temperature. These regressions are evaluated on the clock samples covered by each monitor (Methods).

Retaining the excluded transient excursions gives a lunar response of 0.96 ± 0.14, and resampling the 29 operating groups gives a 95% interval of [0.76, 1.18]. A separate reduction of the 2026 records, using non-overlapping means and ten-day baselines, gives 0.89 ± 0.28 with daily terms. The transient screening removes 2.2 h, or 0.4% of the samples, and its criterion does not use a tidal template. Retaining these excursions changes the lunar central value by only 0.01. In the held-out comparison, the amplitude estimated from the other periods predicts the excluded period, with only its segment offsets fitted. Lunar predictions improve six of eight periods, testing the response in observations excluded from the amplitude fit. Agreement across these choices supports a lunar response that persists across observation periods and sample selections. Methods specifies the samples and uncertainty procedures for these checks.

Discussion

Distant optical clocks resolve time-dependent tidal potential into lunar and solar-frequency semidiurnal contributions. The equivalent-height spectral densities, conditional scans, reciprocal subtraction and correlation analysis provide mutually supporting evidence for their separate detection. The grouped responses and lunar phase agree with the combined solid-Earth and ocean-loading prediction. The comparison extends clock-based potential measurements to resolved temporal structure. Grouped fitting relates the observations to the physical tide prediction, the phase comparison checks the timing of the lunar response, and the spectra expose the contributions at neighbouring frequencies. The separation is supported by the response retained after reciprocal subtraction and by the measured coupling of the two components. These results establish the information about potential dynamics available from the interrupted comparison.

Phase coverage, clock stability and observation time determine how precisely the com-

ponents can be measured. The summer record shows the value of covering the lunar–solar phase cycle, while the campaign-dependent daily fits quantify the remaining sensitivity to changing laboratory conditions. Coastal-model sensitivity, which shifts a fitted response by approximately 0.06–0.07 in the present tests, also matters as the statistical uncertainty decreases. Extending the observation time can improve both the accumulated frequency precision and the range of relative tidal phases sampled. Improved clock stability reduces the uncertainty within each observing interval. Noise limits the precision of each component, while phase coverage and gaps determine how well neighbouring responses can be distinguished. Comparisons across seasons would also allow the stability of those responses to be examined under changing environmental conditions.

A contemporaneous independent observation of tidal potential variations in a European optical-clock network has also been reported.[24]

Methods

Clocks, frequency transfer and sampling

The Sr optical lattice clock is located in Shanghai (31.130° N, 121.570° E, height approximately 5 m) and the Yb optical lattice clock in Wuhan (30.529° N, 114.357° E, approximately 30 m). The tidal model is insensitive to the absolute heights: changing the Shanghai height by 5 m changes the differential model by 4×10^{-23}. The geopotential difference between the clocks, 280.042 ± 0.085 m^2 s^{-2}, was determined by first-order levelling. It enters the static frequency ratio and is absorbed by the fitted offsets. The evaluated systematic uncertainties of the Sr and Yb clocks are 9.2×10^{-19} and 1.3×10^{-18}, respectively [25,26]; optical frequency dissemination over field fibres of comparable length has also been demonstrated [27,28].

A 1,550-nm carrier referenced to the Sr clock through a frequency comb in Shanghai is transmitted to Wuhan, where its beat against the Yb-referenced comb is recorded at 1-s intervals. In the 2026 campaigns, the beat frequency is 33.6 MHz and a deviation of 1 Hz corresponds to 5.17×10^{-15} in Yb/Sr. In 2025, the counter recorded the beat after division by two; these deviations are therefore multiplied by two. Fitting the September–October 2025 campaign alone gives $A = 1.28 \pm 0.31$, consistent with this scale and sign. The white-frequency-noise level of the tide-corrected ratio, $\sigma_y\sqrt{\tau}$, obtained from the overlapping Allan deviation [29] at averaging times of 256–4,096 s, is 7.9×10^{-16} in September–October 2025 and 4.2–5.3×10^{-16} in the 2026 campaigns.

The record contains five campaigns: September–October 2025, January 2026, April 2026, May 2026 and June–August 2026. The 54 segments were defined by the experiment team. There are 38 segments and 270 h before June and 16 segments and 274 h in June–August, giving 544 h, or approximately 23 days, of retained data. The analysis distinguishes these five campaigns from eight observation periods and 29 operating groups. Campaigns define the seasonal subsets and campaign-specific daily terms; observation periods are the units of held-out prediction and period-level sign resampling; operating groups are the units of group bootstrap resampling; segments carry the offsets in the

primary fit. These levels are not treated as interchangeable independent samples.

Data selection

The same selection procedure is applied to every segment. Within each supplied segment, times are assigned from the first file timestamp and the nominal 1-s sample index. This assignment assumes uninterrupted acquisition within each retained run. Samples more than 10 Hz from the segment median are removed, the longest uninterrupted run is retained, and leading and trailing samples deviating from the median by more than 1% of the peak-to-peak range are trimmed. The 10-Hz threshold corresponds to approximately 5×10^{-14} in Yb/Sr. To identify smaller transient excursions, the retained 1-s data are averaged over 300 s and compared with a running median over ±30 min. Averages deviating by more than six robust standard deviations of these differences within the segment are removed together with their neighbours. This step removes 0.4% of the samples, or 2.2 h, including an excursion of approximately -2×10^{-16} lasting about 20 min on 7 July 2026.

The transient criterion uses no tidal model. Adding the model with factors of -1, 1 or 3 leaves the removed samples unchanged. Six robust standard deviations is the lowest tested threshold for which this invariance holds up to three times the model. For block lengths of 60 or 300 s and thresholds from four to eight standard deviations, the lunar amplitude ranges from 0.93 to 1.02. Retaining the excursions gives 0.96 ± 0.14; the full-template and semidiurnal amplitudes change by less than 0.04. After removal, the campaign diurnal amplitudes are mutually consistent ($p = 0.21$). Because the tested tidal additions do not change selection and the subsequent filter is linear, the injection simulations use the selected data.

Durations refer to retained 1-s samples; exposure measured from the filtered data at 600-s steps, used for temperature overlap, is slightly shorter. The last April segment ends at 18:00 on 19 April, following the experiment team's segment table. Omitting endpoint trimming while retaining the other steps changes the lunar amplitude by -0.009 and the full-template amplitude by 0.008. Each segment has its own fitted offset. Time stamps are in local time (UTC+8); shifting the model by ±10 min changes A by less than 0.02. Long-period tides are outside the analysis: the similar site latitudes make the predicted differential semiannual tide only about 2×10^{-20}.

Solid Earth tide and ocean loading

The solid Earth contribution at each clock is computed directly as

$$\delta W = \sum_m (1 + k_{2m}) V_{2m} + (1 + k_3) V_3 - g u_r .$$

Here V_{2m} are the degree-2 tide-generating potentials of the Moon and Sun separated by order m, V_3 is the lunar degree-3 potential, and u_r is radial displacement. We use $k_{20} = 0.29525$, $k_{22} = 0.29801$ and $k_3 = 0.093$. The IERS 2010 resonance formula

is applied constituent by constituent for k_{21}, giving 0.2588 for K_1, 0.2873 for P_1 and 0.2975 for O_1. The radial displacement is computed with the IERS algorithm, including the frequency and latitude dependence of the Love number h [20].

The ocean-loading potential is computed by grid convolution:

$$\delta W_{\mathrm{ocean}}(t) = \sum_{\mathrm{ocean}} \rho_w \zeta(\mathbf{r}', t) G_W(\psi)\, \mathrm{d}A, \qquad G_W(\psi) = \frac{ga}{M_E} \sum_n (1 + k'_n - h'_n) P_n(\cos\psi).$$

The ocean elevation ζ is taken from FES2022 on a $1/30^\circ$ grid [21,22], $\rho_w = 1,025\ \mathrm{kg\,m^{-3}}$, a and M_E are Earth's radius and mass, and ψ is angular distance. The Green's function follows ref. [23]. PREM load Love numbers are used through degree $n = 696$ [30], from the gravity-toolkit distribution [31]. Above this degree, h'_n is joined smoothly to its Boussinesq limit of -6.22, and the series is summed with a Kummer transformation through $n = 40,000$. The convolution uses full grid resolution within 3° of each station and sums blocks of 8×8 cells farther away. This aggregation changes differential M_2 and K_1 loading by at most 0.05%. The ocean mean is removed to enforce mass conservation. Eight constituents—Q_1, O_1, P_1, K_1, N_2, M_2, S_2 and K_2—are synthesized with astronomical arguments and nodal corrections.

The full loading potential exceeds the vertical-displacement contribution $-g\delta h$ by factors of approximately 1.4 for the semidiurnal constituents and 1.8 for the diurnal constituents. The radial displacement from the same convolution agrees with the FES2022 load-tide grid to within 2% in amplitude and 1.2° in phase for seven of the eight constituents. This comparison checks the convolution and displacement contribution; the direct attraction and induced-potential terms use the same PREM load Love numbers. For S_2, the amplitude differs by 18% and the phase at Wuhan by 54°; the load-tide grid uses a different ocean-model release. The S_2 loading contributions expressed as fractional frequency are 0.62×10^{-18} in Shanghai and 0.18×10^{-18} in Wuhan, giving a differential amplitude of 0.63×10^{-18}, compared with 1.42×10^{-18} for M_2. Substituting the S_2 loading inferred from the load-tide grid changes the full-template amplitude by 0.001 and the semidiurnal amplitude by 0.000 at the reported precision, with no reported change in the lunar amplitude. The lunar and near-12-h groups have separate fitted amplitudes.

The Shanghai clock is 22 km from the nearest ocean cell. To examine coastal-mask sensitivity, land cells within 0.3° or 0.6° of the coastline in a $4^\circ \times 5^\circ$ box centred on Shanghai are filled with the nearest ocean values. These strips, approximately 33–66 km wide, increase the Shanghai M_2 loading potential expressed as $\delta W/g$ from 15.3 mm to 17.0 or 18.5 mm and decrease the fitted response by 0.06 or 0.07. These are physical-model sensitivity tests, with shifts approximately half the quoted lunar statistical uncertainty; they are not treated as independent Gaussian errors or as an exhaustive loading-model uncertainty.

Tidal fits and uncertainty

The 1-s observations and model are filtered identically using a 1,200-s triangular window stepped by 600 s. The whole-template model is fitted by least squares as $y(t) = c_i + As(t)$, with an offset c_i for segment i. For the band analysis, the eight-constituent model is divided either into semidiurnal (M_2, S_2, N_2, K_2) and diurnal (K_1, O_1, P_1, Q_1) bands, or into lunar (M_2, N_2), near-12-h (S_2, K_2) and diurnal groups with separate amplitudes. The primary grouped fit contains no empirical daily terms. The full-template response is 0.90 ± 0.12 and the jointly fitted diurnal response is 0.35 ± 0.51. A separate loading coefficient is 0.89 ± 0.55. If the full physical model is correct, fitting the solid Earth tide alone on the actual sampling is expected to give $A = 1.32$, because loading adds to the solid tide with an M_2 phase difference of approximately 20°.

Where indicated, uncertainty is the larger of the scaled least-squares standard error[32] and the day-by-day jackknife error. For the full template, semidiurnal band and lunar group, uncertainty is calibrated by residual resampling. The fitted residuals are shifted circularly within each continuous run of the filtered record and added to the model with $A = 1$; the standard deviation of the refitted amplitude over 200 realizations supplies the calibrated error. These shifts retain the within-run residual structure under circular wrapping. For the lunar group, two independent sets of 200 realizations give standard deviations of 0.13 and 0.14, compared with 0.13 for the larger of the scaled least-squares and jackknife errors. We quote 0.14; the resampling bias is +0.01. With the daily disturbances described below, 5,000 realizations per scenario give standard deviations rounding to 0.14, coverage of 0.67–0.68 for the interval ±0.14, and 0.945–0.952 for ±1.96 × 0.14.

Calibrated errors are used for the whole-template and semidiurnal responses of individual campaigns and of the full record. For the diurnal and near-12-h groups, fits containing empirical daily terms, and fitted daily amplitudes, the larger of the scaled least-squares and jackknife errors is used. In the resampled data, these errors differ from the calibrated uncertainty by factors of 0.8–1.3. Unless otherwise specified, quoted ± uncertainties denote one standard deviation or one standard error as defined by the applicable procedure.

Group-bootstrap intervals are obtained by drawing the 29 operating groups with replacement and refitting 1,000 times; the segment bootstrap instead draws the 54 segments. These procedures permit correlations within each resampling unit but assume independence between units. For the semidiurnal response, the group-bootstrap standard deviation of 0.08 is smaller than the calibrated value of 0.12, and the larger value is quoted. In the phase test, each tidal band has an in-phase and a quadrature template, the latter formed by shifting every constituent by 90°. Segment-bootstrap standard deviations give a lunar amplitude of 0.98 ± 0.11 and phase of $-6^\circ \pm 8^\circ$.

The phase-folded summer display is formed after subtracting the fitted segment offsets and non-lunar groups and then averaging in 30° M_2-phase bins; the M_2+N_2 model is folded with the same sampling. Its day-by-day jackknife error bars hold the fitted non-lunar terms fixed. They are conditional display uncertainties and do not propagate un-

certainty in those fitted terms; they are distinct from the uncertainty of the joint lunar-amplitude estimate. The summer time-series display uses 3-h averages and removes gaps longer than 6 h from its horizontal axis, marking their locations.

The eight constituents explain 99.84% of the model variance. The remaining component, with an r.m.s. of 0.18×10^{-18}, is retained at its predicted value. Setting it to zero or doubling it gives semidiurnal amplitudes of 0.95–0.96 and lunar amplitudes of 0.95–0.99. An additional central-value check uses a Kalman filter with state (segment offset, A), resetting the offset at each segment and allowing a random walk reaching 5×10^{-18} after 6 h. It gives $A = 0.95$ for the full record; changing that scale to 2×10^{-18} or 10^{-17} gives 0.92 or 0.99, respectively.

Daily disturbances and validation

Daily sensitivity fits add terms $a_1 \cos \omega t + b_1 \sin \omega t + a_2 \cos 2\omega t + b_2 \sin 2\omega t$, where $\omega = 2\pi/(1\ \text{day})$ and t is local time. Common coefficients and campaign-specific coefficients are examined. A physically constrained alternative uses a free-amplitude 24-h term in each campaign with the phase of the 24-h harmonic of hourly ERA5 2-m air temperature at the two sites [33], averaged over that campaign. Another alternative includes the air-temperature anomalies at both sites relative to a 24-h running mean as two regressors. Within a short campaign, a 24-h term is partly degenerate with diurnal tides and segment offsets.

The sensitivity of the lunar estimate to $B \cos(\omega t - \varphi)$ is evaluated using the lunar row h of the least-squares operator. For a common phase, the maximum amplitude change is

$$B\sqrt{(h \cdot \cos \omega t)^2 + (h \cdot \sin \omega t)^2}.$$

For phases independent between campaigns, the maximum is the sum of the corresponding campaign norms. These bounds describe the specified sinusoidal disturbances and phase assumptions; they do not exclude arbitrary time-varying daily effects.

Injection tests use circularly shifted residuals of the full-template fit, adding either the full tidal model ($A = 1$) or the model without its lunar group ($A = 0$). Four daily-disturbance scenarios are considered: no added daily disturbance; campaign-specific 24-h terms with air-temperature phases and amplitudes fitted to the observations; common 24-h and 12-h terms with fitted amplitudes and phases; and the fitted summer 24-h and 12-h terms. The added disturbances have r.m.s. values up to 1.9×10^{-18}. In 300 realizations per scenario and injected amplitude, lunar-amplitude biases lie between -0.03 and -0.00 at the reported precision. The larger of the scaled least-squares and jackknife errors covers the injected value in 0.59–0.71 of realizations at one standard error and 0.90–0.95 at 1.96 standard errors. The main estimate uses the calibrated uncertainty above. For $A = 1$, the fitted lunar amplitude exceeds three times its estimated error in every realization; for $A = 0$, this fraction is at most 0.003. These simulation results are conditional on the selected residual and disturbance models.

A complementary restricted wild bootstrap flips the residual sign of whole observation

periods under the null. With eight periods, the smallest attainable two-sided probability is $2/256$. The resulting two-sided probabilities are 0.008 for the lunar estimate and 0.008 for the whole-template estimate. Campaign information shares are calculated by projecting the lunar template onto the complement of the other fit columns—near-12-h and diurnal templates and segment offsets—and summing its squared projected values within each campaign.

Held-out prediction estimates the amplitude from the other observation periods and applies it to the excluded period, fitting that period's segment offsets. The whole-template prediction reduces residual sums of squares in seven of eight periods, by 4.9% when pooled; the lunar prediction improves six of eight periods, by 1.7% when pooled.

Alternative processing

Independent processing of the same 2026 records uses non-overlapping 600-s means, corrected clock systematic shifts and fixed ten-day calendar baselines. It gives a whole-template response of 0.88 ± 0.10 and a lunar response of 0.89 ± 0.28 when daily terms are included in each ten-day block. Its whole-template prediction improves all six held-out 2026 periods. Simulations with white, Ornstein–Uhlenbeck (correlation times of 1 h to 3 d), Student-t and moving-average noise give biases below 0.04 and nominal 95% interval coverage of 94.5–99.8%, using the largest of least-squares, delete-day and delete-period standard errors. Simpler solid-tide approximations change its whole-template scale by less than 0.01. These estimates are processing checks on shared observations and are not combined as independent measurements.

Fibre-link monitor

The round-trip beat of the fibre link is recorded at the Shanghai end near 10 MHz with 1-s resolution. Its deviations are normalized by the optical carrier frequency without division by two and filtered in the same way as the clock ratio. Samples more than 10 Hz from the median are excluded. Monitor excursions reaching 10^{-12} leave the recorded ratio unchanged. Additionally excluding 300-s monitor blocks deviating by more than 5×10^{-16} changes the regression result by less than 0.005. The monitor covers 474 h of the analysed data. On that overlap, the lunar and whole-template amplitudes are 0.93 and 0.80; including the monitor as an additional regressor changes each by less than 0.005.

Environmental analysis

The Sr vacuum-chamber temperature is the mean of 20 PT100 sensors sampled every 20 s, available from 28 June to 7 July 2026 and overlapping 98 h of the filtered summer record. The blackbody sensitivity is 6.26×10^{-17} K^{-1} at 294.3 K; a temperature increase raises Yb/Sr. The daily temperature profile is obtained after subtracting a 24-h running mean and excluding the chamber-temperature transient on 30 June. Both clocks apply blackbody corrections in real time, so the recorded ratio already includes them. The

regression therefore tests correction errors proportional to chamber temperature; its expected coefficient is zero.

Over the 98-h overlap, the blackbody shift calculated from chamber temperature and filtered like the ratio has an r.m.s. of 2.8×10^{-18}, including the 30 June transient. Adding it with a free coefficient gives 0.92 ± 0.74, or 0.72 ± 0.74 after excluding the transient, with day-by-day jackknife errors. Both are consistent with zero at this precision. Through the summer lunar-estimator response of 0.016 per 10^{-18}, a residual with the size of the observed 24-h ratio term would change the lunar amplitude by about 0.045. Without real-time correction, the blackbody shift would contain a daily component of 2.0×10^{-18} peaking at 12:13 local time. In the same overlap, the fitted 24-h ratio term is $(0.9 \pm 1.1) \times 10^{-18}$ with maximum at 10:54 ± 5.0 h. The mean daily profile of the uncorrected blackbody contribution has a peak-to-peak range of 4×10^{-18}, indicating the size of the removed effect.

Equivalent-height power spectral density

The separate non-overlapping reduction contains 3,299 means and 545.924 member-hours in 54 segments. We use its recorded fractional-frequency means, expressed numerically in units of 10^{-18}, at their stored UTC bin centres. The trial sinusoids are evaluated at these centres, following the bin-centre spectral convention; within-bin member integration and the main analysis's triangular filter are not reconstructed here. Equal spectral weights differ from the noise-based weights used for the physical amplitude estimates in that reduction.

We express fractional-frequency variations y as an equivalent height $h_{\mathrm{eq}} = c^2 y / g_0$, with $c = 299,792,458\ \mathrm{m\,s^{-1}}$ and the standard-gravity convention $g_0 = 9.80665\ \mathrm{m\,s^{-2}}$. This is the height equivalent of the measured potential variation, including attraction and deformation, rather than a measurement of vertical displacement. This reporting convention leaves the gravity values in the physical tide model unchanged.

For a nuisance matrix N, we form its orthonormal basis Q and apply $R = I - QQ^{\mathrm{T}}$ to the observation vector and each trial cosine–sine pair. The score ΔRSS is the reduction in residual sum of squares when the projected pair is fitted. With y expressed numerically in units of 10^{-18}, our nominal-600-s normalization is

$$D_h(f) = \left(\frac{c^2}{g_0}\right)^2 (600 \times 10^{-42})\, \Delta\mathrm{RSS}(f) \quad \mathrm{m^2/\mu Hz}.$$

The factor 10^{-42} combines the squared fractional-frequency scale, 10^{-36}, with the conversion from density per Hz to density per µHz, 10^{-6}. Equivalently, each unit of numerical ΔRSS contributes $5.039561053 \times 10^{-8}\ \mathrm{m^2/\mu Hz}$. This normalization gives a common scale for the gapped observations and all subtraction and injection curves. Its frequency integral is not generally the time-domain variance after nuisance projection on irregular sampling. All density axes are linear.

The reported M_2 and S_2 densities are evaluated at their exact astronomical frequencies,

22.3642779633 and 23.1481481481 µHz. Both conditional estimates control 54 segment constants, free N_2 and K_2 pairs, and the other principal pair. Their values are $1.279189285 \times 10^{-3}$ and $1.936689191 \times 10^{-4}$ m^2/μHz, respectively (Fig. 3d). With only the segment and N_2/K_2 terms controlled, the corresponding densities are $1.556090557 \times 10^{-3}$ and $4.705701906 \times 10^{-4}$ m^2/μHz (Fig. 3a). These are fixed-frequency ordinates, distinct from the maxima of a local scan. The density values are point estimates; uncertainty in the underlying sinusoidal coefficients is summarized by the joint regions below. Extended Data Fig. 7a uses segment constants alone; its other panels also control N_2 and K_2. All phases refer to 6 September 2025, 16:05 UTC, the first stored bin centre.

The M_2 and S_2 subtraction waveforms come from a simultaneous four-coefficient fit after the same nuisance projection. Conditional scans add the other principal pair to N and scan within ±0.16 µHz of the target. No physical amplitude ratio or phase constraint links the two principal components. As a nuisance sensitivity check, their joint amplitudes without N_2/K_2 control are 4.63 and 1.82 × 10^{-18}. Scaled least-squares and UTC-day-deletion covariances quantify the joint coefficient uncertainties (Extended Data Table 4 and Extended Data Fig. 9). Synthetic injection curves contain only the fitted principal sinusoidal waveform evaluated at the measured times, with no added random noise.

Compensation sensitivity and template correlation

For donor D and target T, the joint fit supplied the fixed donor coefficient vector $\hat{\beta}_D$. For 61 values of λ from 0 to 1.5, we formed $y_\lambda = y - \lambda X_D \hat{\beta}_D$ and fitted only the target pair together with N. With $P_T = R_N X_T$ and $P_D = R_N X_D$, its coefficient vector obeys

$$\hat{\beta}_T(\lambda) = \hat{\beta}_{T,\text{single}} - \lambda P_T^{+} P_D \hat{\beta}_D.$$

We also repeated the sweep using the donor's single-frequency estimate to show the dependence on waveform definition. A separate jointly controlled fit retains both M_2 and S_2 pairs after subtraction. Its target coefficient is necessarily invariant, because the subtracted waveform lies within the fitted donor subspace; this curve is an algebraic control, not evidence against leakage. Amplitude is the norm of the two coefficients and phase is $\text{atan2}(\beta_{\text{sin}}, \beta_{\text{cos}})$, corresponding to $A\cos(2\pi f(t - t_0) - \phi)$. Target-only incremental improvement is $\Delta\text{RSS} = \hat{\beta}_T^{\mathsf{T}} P_T^{\mathsf{T}} P_T \hat{\beta}_T$. The jointly controlled improvement instead residualizes the target against both N and the donor, so these two improvements have different baselines even where their coefficient estimates coincide. Density displays use the equivalent-height normalization defined above.

Canonical correlations are the singular values of $Q_M^{\mathsf{T}} Q_S$, where each Q spans a residualized constituent pair. Template lag correlations evaluate the fitted donor waveform at $t - \ell$ and the target at t, residualize both on the unchanged observation support, and normalize their inner product. Lags span ±24 h in 5-min steps. Their zero-lag correlation is 0.1955; maximum absolute lag correlations are 0.2413 for shifted M_2 against S_2 and 0.2113 in the reverse direction. These correlations describe the design and fitted-

template similarity on the observation support.

For alias diagnostics, we scanned 20–25 μHz with spacing $1/(12T)$, where T is the elapsed observation span. For each donor, we selected the three strongest local maxima of its pure-injection spectrum at least $2/T$ away from its own frequency. Selection used the injection response, not a significance threshold on the observed data. At each selected frequency, we compared the actual refit to the predicted vector response $\widehat{b}_f(\lambda) = \widehat{b}_f(0) - \lambda\widehat{b}_{f,D}$. Its squared norm in the frequency-specific Gram metric gives the predicted quadratic power response. The observed peak locations and their compensation minima are compared with this sampling response. The compensation curves describe deterministic sensitivity conditional on the fitted waveforms.

We express each four-coefficient covariance as a correlation matrix, $R_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$, separately for scaled least squares and UTC-day deletion (Extended Data Fig. 9c,d). These matrices describe the coupled uncertainty of the joint coefficient estimates, distinct from correlations of the design columns and fitted templates.

Joint coefficient regions

For each constituent, we took its cosine–sine coefficient vector $\hat{\beta}$ and the associated 2×2 block C of the joint-fit covariance. We formed the nominal 95% region $(\beta - \hat{\beta})^{\mathsf{T}} C^{-1} (\beta - \hat{\beta}) \leq 5.991$, using the two-degree-of-freedom chi-square quantile (radius $\sqrt{5.991}$). We calculated the extrema of $\|\beta\|$ over this region; the lower bound is zero if the origin is inside, otherwise it is the minimum norm on the ellipse boundary. Phase is $\mathrm{atan2}(\beta_{\sin}, \beta_{\cos})$, with time origin 6 September 2025 at 16:05 UTC; phase extrema are reported only when the region excludes the origin. The ranges are projections of a joint coefficient region, not symmetric amplitude standard errors. Coverage is nominal under the normal approximation, separately for each constituent. The 43 UTC dates are the jackknife deletion blocks. The UTC-day covariance gives amplitude ranges of $[2.94, 6.09] \times 10^{-18}$ for M_2 and $[0, 4.42] \times 10^{-18}$ for S_2. The S_2 region includes zero and does not constrain its phase at this level. Scaled least-squares regions provide a comparison using a second covariance estimate.

Observation support of the two reductions

The two reductions use overlapping records from 54 acquisition segments. Their member exposure, averaging and time conventions differ.

Definition	Main reduction	Auxiliary spectral reduction
Selection	Segment screening, endpoint trimming and transient exclusion	Existing segment cuts and jump screening; no main-route transient exclusion

Definition	Main reduction	Auxiliary spectral reduction
Retained exposure	Approximately 544 h of nominal 1-s observations; transient exclusion removes 2.2 h	1,965,327 nominal 1-s members, or 545.924 member-hours
Averaging	1,200-s triangular filter, stepped by 600 s	Non-overlapping 600-s wall-clock bins, retaining at least 180 members
Output observations	Filtered centres at 600-s steps	3,299 means; 3,291 distinct bin centres, with shared centres in different segments
Estimation	Least squares with residual-calibrated uncertainty	Equal row weights for the spectra and free-phase harmonic fits
Reference levels	One fitted constant per segment	The same 54 segment indicators
Time assignment	First file timestamp plus nominal 1-s sample index; local time is UTC+8	Stored UTC bin centres

The auxiliary bin ledger contains 1,966,908 members before the minimum-count rule; rejecting 1,581 members in 29 underfilled bins leaves 1,965,327. The two exposures also reflect different endpoint, timing and averaging rules; their approximate 1.9-h difference is not a direct measure of transient exclusion.

Author contributions

Jian-Wei Pan planned and directed the study and contributed to manuscript writing. Hai-Feng Jiang designed and coordinated the optical-clock comparison and organized its implementation. Qiang Zhang designed and coordinated the comparison experiment. Lei Hou coordinated the remote frequency comparison and was responsible for the optical frequency combs at both stations. Xiang-Pei Liu designed and coordinated the Sr-clock comparison and evaluated its gravitational redshift. Qi Shen designed the optical-frequency-comb and clock-comparison experimental schemes. Sheng-Kai Liao, Juan Yin and Cheng-Zhi Peng contributed to discussions of the experimental design and planning.

Han-Ning Dai was responsible for the overall Sr-clock design. Xiang Zhang operated and maintained Sr1 and analysed the Sr-clock data. Jia-Hao Zhang and Long-Fei Liao operated and maintained the Sr clock. Xiao Jiang and Yu-Ao Chen were responsible for the Sr-clock system. Yu-Chen Zhang was responsible for the link between the Sr1 ultrastable laser and the frequency comb and for height measurements in Shanghai. De-Quan Kong maintained the U1 ultrastable laser. Xing-Yang Cui constructed and maintained the Sr-clock laser servo systems; Ping Xu constructed and maintained its ultrastable

clock-laser system; Peng Liu constructed and maintained the atomic-physics system; and Jie Li constructed and maintained its electronics. Hua-Qing Zhang optimized and modified the Sr-clock laser output.

Zhuan-Xian Xiong and Baolong Lyu were responsible for the overall Yb-clock system. Qiang Zhu constructed the Yb-clock optical system, designed its electronics and analysed data. Bing Wang constructed the optical system and operated and maintained the Yb clock; Ling-Xiang He operated and maintained the Yb clock. De-Zhi Xiong performed simulations and analysis of the Yb-clock blackbody-radiation shift and ultrastable-cavity vibrations. Baolong Lyu also performed theoretical analysis and data processing for the AC Stark-shift uncertainty.

Fa-Xi Chen was responsible for the fibre link, and Li-Bo Li maintained it. Wen-Xuan Song operated the Shanghai optical frequency comb and acquired and processed data. Jian-Yu Guan was responsible for the optical frequency combs and comparison-data acquisition and processing. Shao-Mao Wang was responsible for the 1,550-nm ultrastable laser used for frequency transfer in the local frequency-comb system. Youchao Xie, Lei Wang and Wen-Bin Shen evaluated the gravitational potentials in Wuhan and Shanghai. Yu-Ao Chen also processed data.

Lei Hou, Wen-Xuan Song, Jian-Yu Guan, Han-Ning Dai, Qiang Zhang, Yu-Ao Chen, Hai-Feng Jiang and Jian-Wei Pan contributed to manuscript writing.

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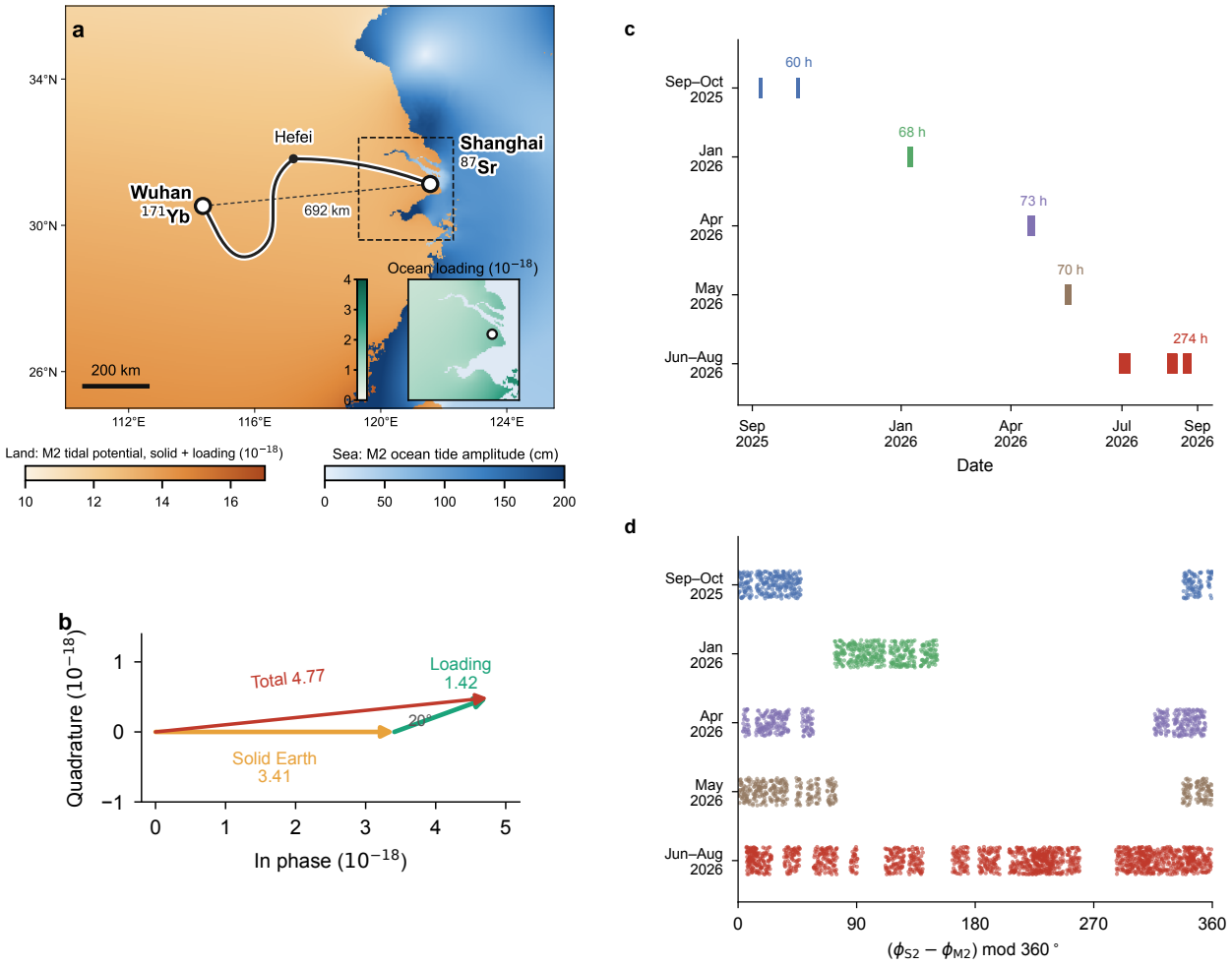


Fig. 1 | The predicted tide on a coastal-inland baseline. **a,** The Sr clock in Shanghai, the Yb clock in Wuhan and the fibre link through Hefei. The dashed line marks the 692-km distance between the clocks. Land colours show the predicted M_2 tidal potential sensed by a clock, including the solid Earth tide and ocean loading, expressed as fractional frequency. The inset shows the ocean-loading contribution around Shanghai; sea colours show the FES2022 M_2 ocean-tide amplitude. **b,** Predicted differential M_2 phasors of the solid Earth tide, the ocean-loading potential and their sum, referenced to the solid Earth tide. **c,** Filtered samples and observation hours in the five campaigns from September 2025 to August 2026. **d,** Relative astronomical phase of S_2 and M_2, including the nodal correction, at the same samples. The short campaigns cover limited relative phases, whereas the summer campaign covers the full cycle as the lunar tide moves through the solar day.

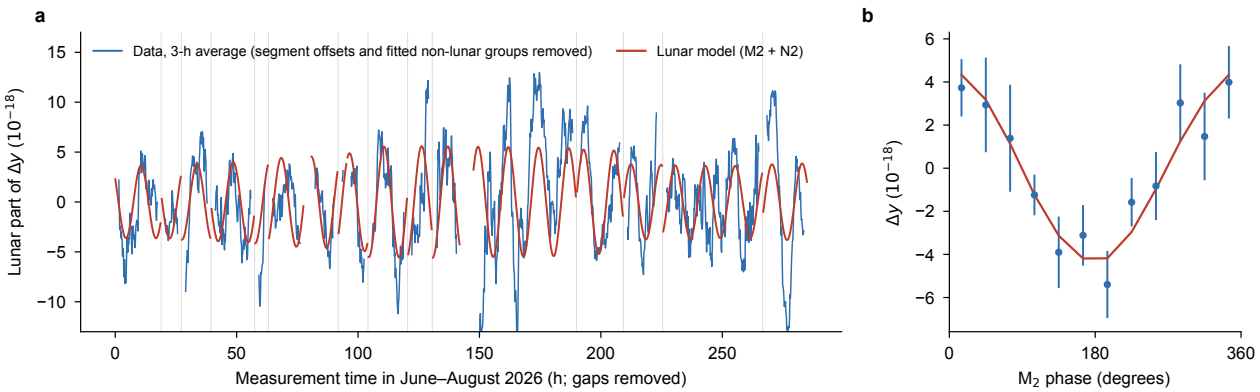


Fig. 2 | The lunar tide in the summer record. a, Three-hour averages of the June–August 2026 frequency ratio after subtracting fitted segment offsets and the fitted near-12-h and diurnal tidal groups (blue), with the lunar model (red). The measurement-time axis compresses gaps longer than 6 h; grey lines mark their removal. White frequency noise alone corresponds to a scatter of approximately 5×10^{-18} for a 3-h average. **b,** The same data folded at the model M_2 phase in 30° bins, with the lunar model (M_2 and N_2) folded using identical sampling. Error bars are day-by-day jackknife uncertainties conditional on the fitted non-lunar terms, which are held fixed during resampling. No empirical daily terms are fitted. Extended Data Fig. 4 provides a complementary view with the actual calendar-time axis.

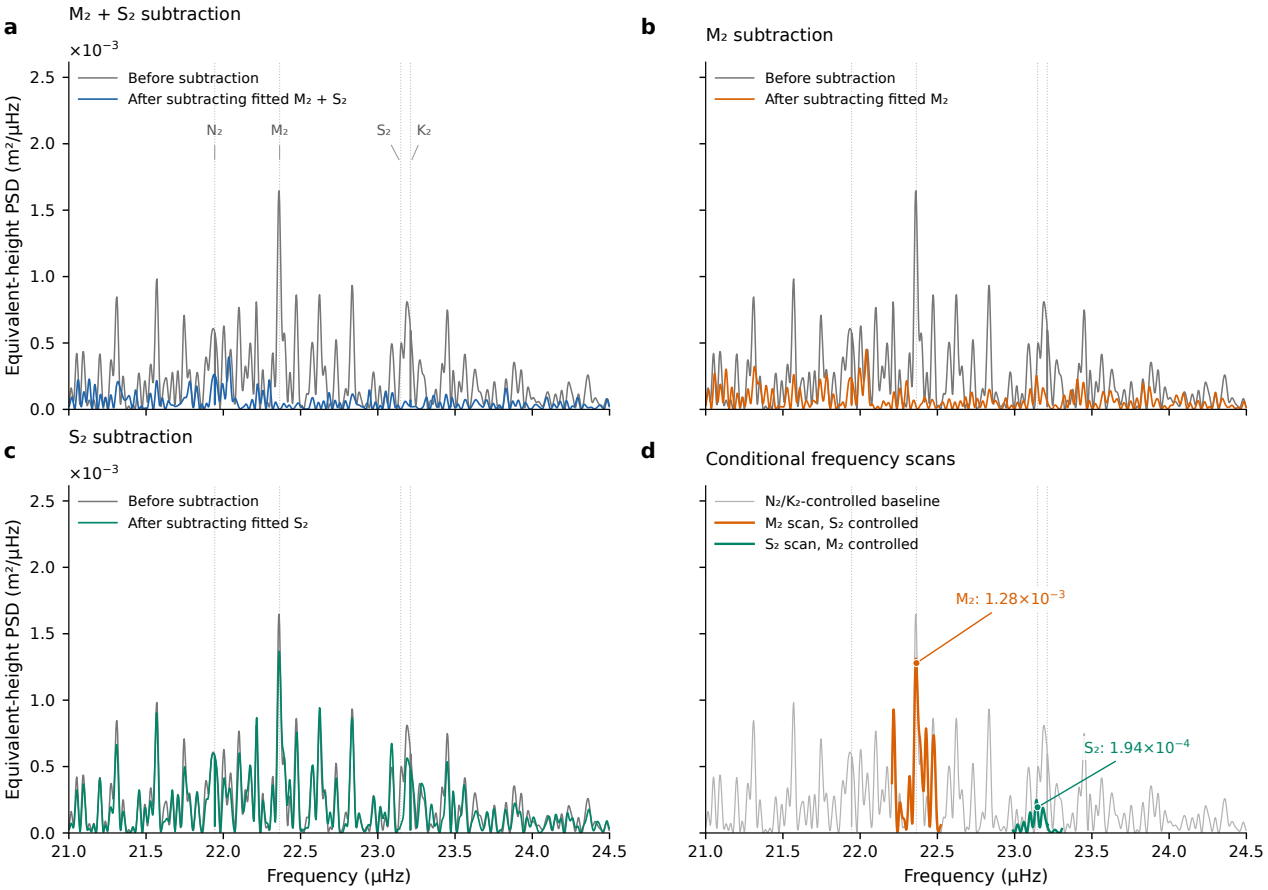


Fig. 3 | Local semidiurnal spectra and component separation. Spectra are calculated from the auxiliary reduction into 3,299 non-overlapping 600-s means at their recorded UTC bin centres, retaining gaps and using equal weights and 54 segment constants. This reduction differs from the triangular filtering used for the main lunar-group amplitude. **a,** Spectra before subtraction (grey) and after subtracting both jointly fitted M_2 and S_2 waveforms (blue). Segment constants and freely phased N_2 and K_2 sine–cosine pairs are controlled in both curves. **b,c,** The N_2/K_2-controlled spectrum before subtraction (grey), after subtracting the fitted M_2 waveform (**b**, orange), and after subtracting the fitted S_2 waveform (**c**, green). Both waveforms are obtained from one simultaneous freely phased M_2–S_2 fit with the same nuisance terms. **d,** Scans within ±0.16 μHz of M_2 with S_2 controlled (orange), and of S_2 with M_2 controlled (green); N_2 and K_2 remain controlled. Both the observations and the trial sine–cosine pair undergo the same nuisance projection. The grey curve is the N_2/K_2-controlled baseline. Dotted lines mark N_2, M_2, S_2 and K_2. All panels show equivalent-height power spectral density in m²/μHz on the same linear scale, using the nominal-600-s normalization in Methods. Markers in d give the values at the exact M_2 and S_2 frequencies: 1.28×10^{-3} and 1.94×10^{-4} $\mathrm{m}^2/\mu\mathrm{Hz}$, respectively. Each controls the other principal pair under the same N_2/K_2 and segment treatment. Extended Data Fig. 7 includes fitted-waveform injections through the same sampling operator to examine sidelobes.

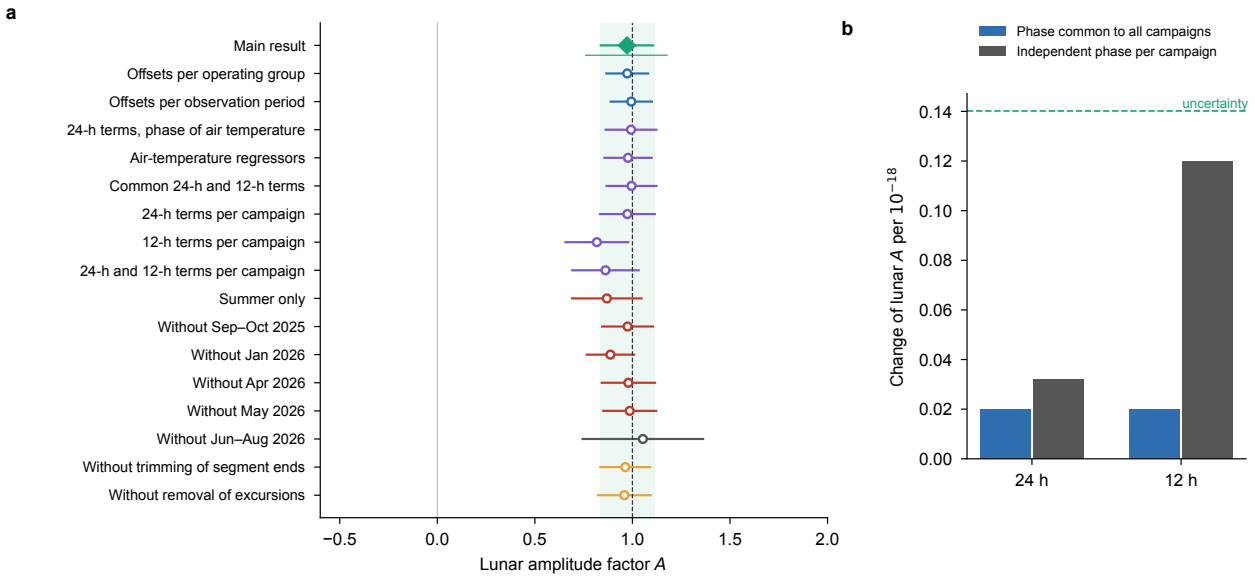
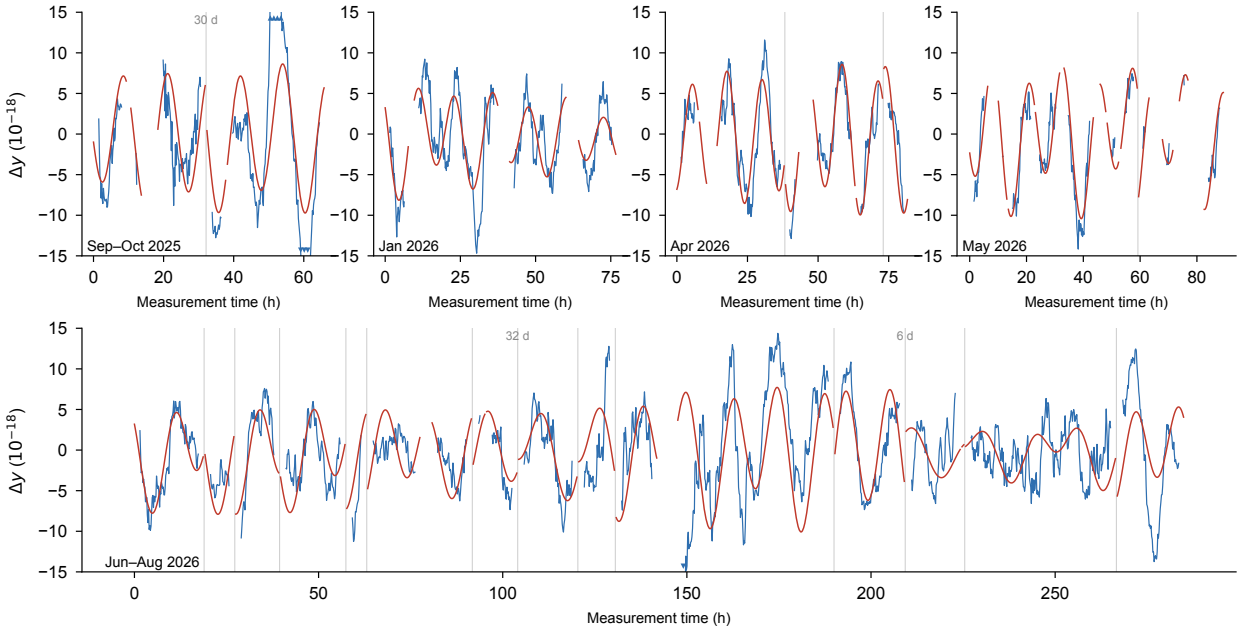
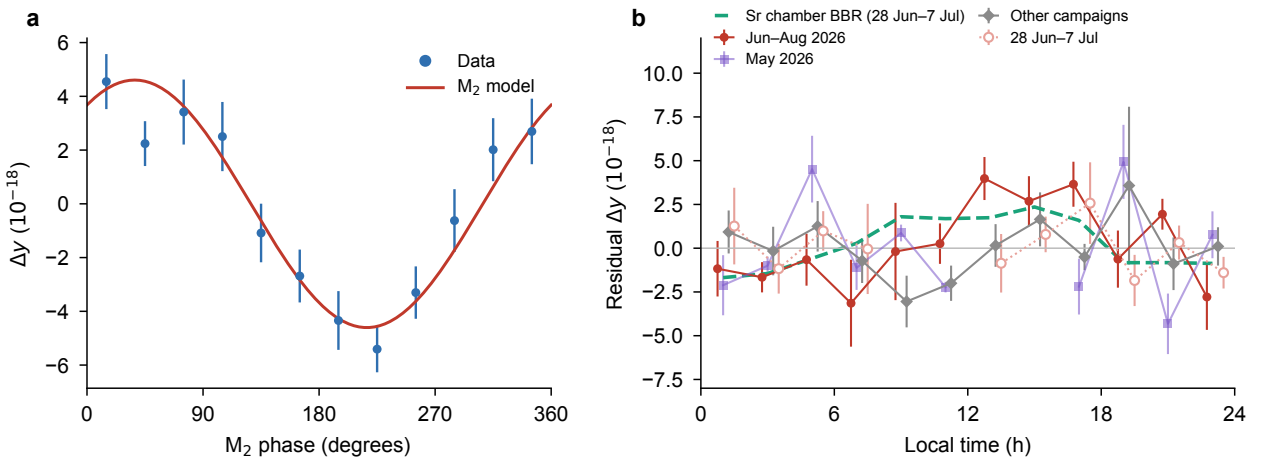


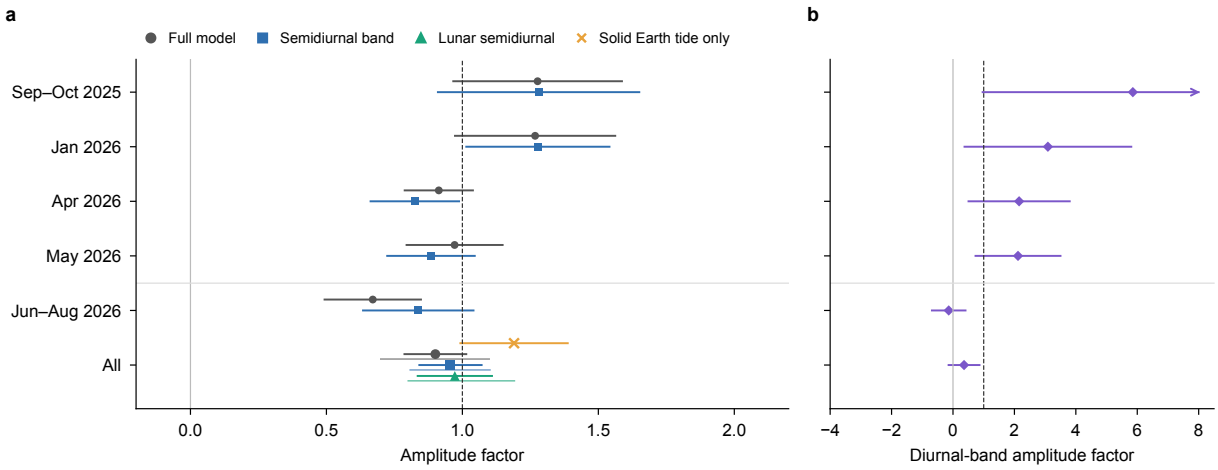
Fig. 4 | Robustness of the lunar amplitude. **a,** Lunar amplitude for the main fit (filled diamond and shaded uncertainty band), alternative offset definitions, daily terms represented by air-temperature measurements or fitted 24-h and 12-h sinusoids, the summer campaign alone, omission of each campaign, and retention of segment ends or transient excursions. The thin bar below the main result is the 95% interval from resampling operating groups. These estimates share observations. The grey interval obtained without the summer campaign shows the loss of information that separates the tidal groups. **b,** Change in the lunar amplitude caused by a 24-h or 12-h sinusoid of fractional-frequency amplitude 10^{-18}, either with one phase common to all campaigns or with the combination of independent campaign phases that produces the largest change. The dashed line marks the uncertainty of the main estimate. The vertical axis is linear.



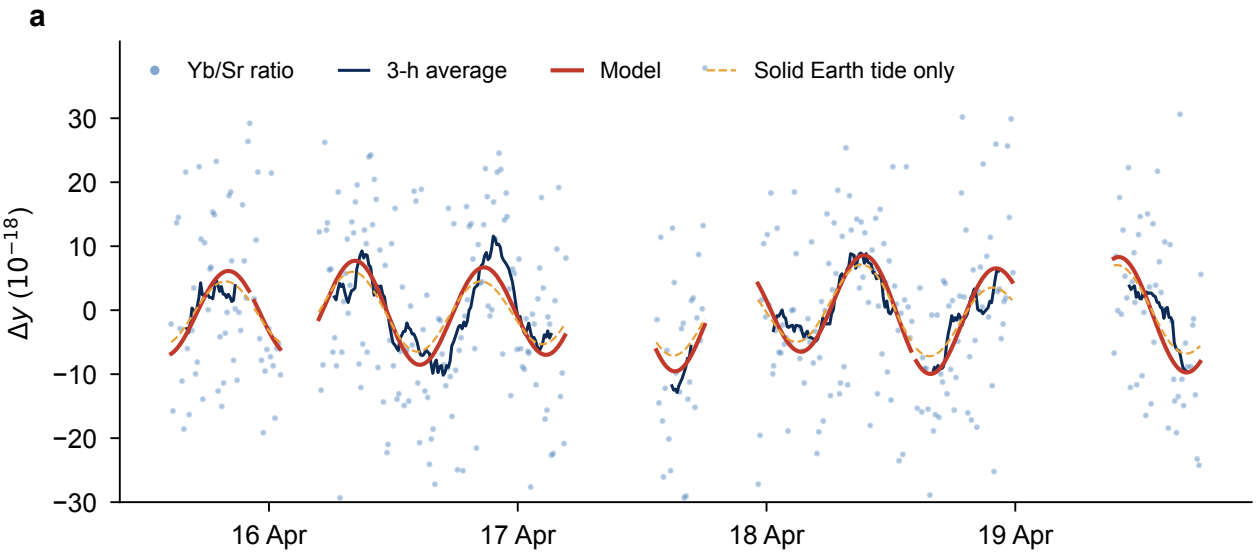
Extended Data Fig. 1 | Tidal records in all campaigns. Three-hour averages of the filtered frequency ratio after subtraction of fitted segment offsets (blue), with the tidal model (red). Measurement-time axes compress gaps longer than 6 h, marked by vertical lines; gaps longer than 5 days are labelled. The vertical range is ±15 × 10^{-18}, and triangles mark values outside this range. The actual calendar-time axis is retained for the April campaign in Extended Data Fig. 4.



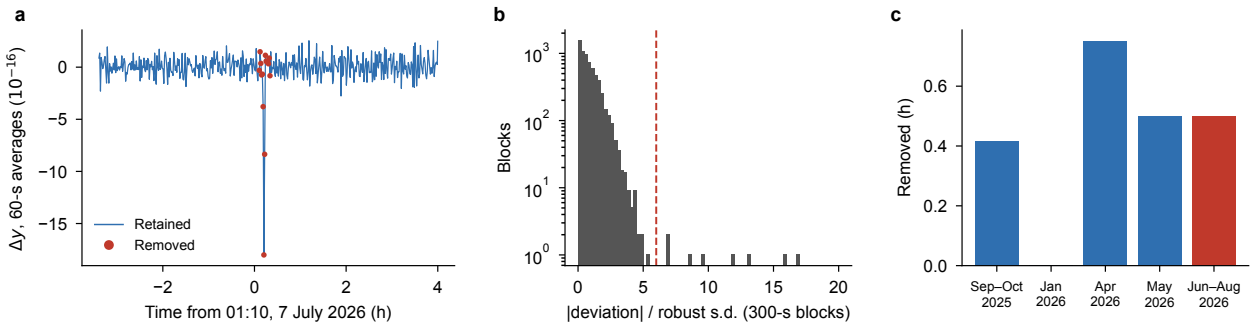
Extended Data Fig. 2 | Folds by tidal phase and local time. a, All observations folded at the M_2 phase after subtracting fitted segment offsets and all tidal constituents except M_2 from the full-model fit. Points are means in 30° bins with day-by-day jackknife uncertainties; the curve is the M_2 model. These phase-bin uncertainties are conditional on the fitted terms removed before folding, which are held fixed. **b,** Tidally corrected residuals folded by local time in 2-h bins for summer, May (light symbols), the other three campaigns, and summer observations overlapping the temperature record from 28 June to 7 July (open circles). Error bars are day-by-day jackknife uncertainties; bins covering fewer than three days are omitted. The dashed curve is the mean daily blackbody-radiation shift computed from the Sr chamber temperature from 28 June to 7 July 2026 after removing a 24-h running mean and excluding the temperature transient on 30 June. The real-time correction removes this chamber-temperature shift from the recorded frequency ratio.



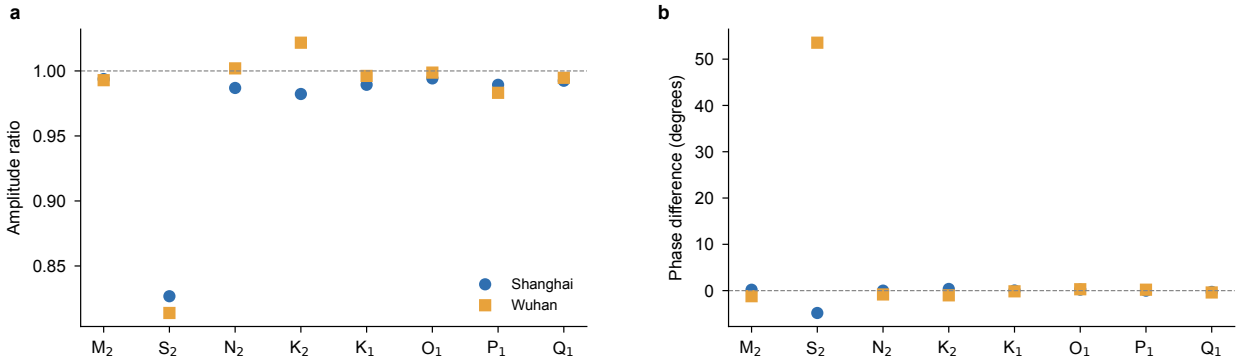
Extended Data Fig. 3 | Campaign-resolved tidal amplitudes. a, Full-model and semidiurnal amplitude factors for each campaign and the combined observations, fitted with segment offsets. Combined-data estimates also include the lunar group and the solid Earth tide alone. Short campaigns constrain the semidiurnal band as a whole; the joint fit separates the lunar group. **b,** Diurnal amplitude fitted jointly with the semidiurnal band in a two-band fit; the combined estimate is 0.36 ± 0.50. The arrow marks an uncertainty extending beyond the plotted range. Error bars represent one standard deviation, calibrated by residual resampling for the full model, semidiurnal band and lunar group; for the diurnal band and solid Earth tide alone they are the larger of the scaled least-squares and day-by-day jackknife uncertainties. Thin lines in the combined-data rows show 95% intervals from resampling operating groups, or segments for the lunar group.



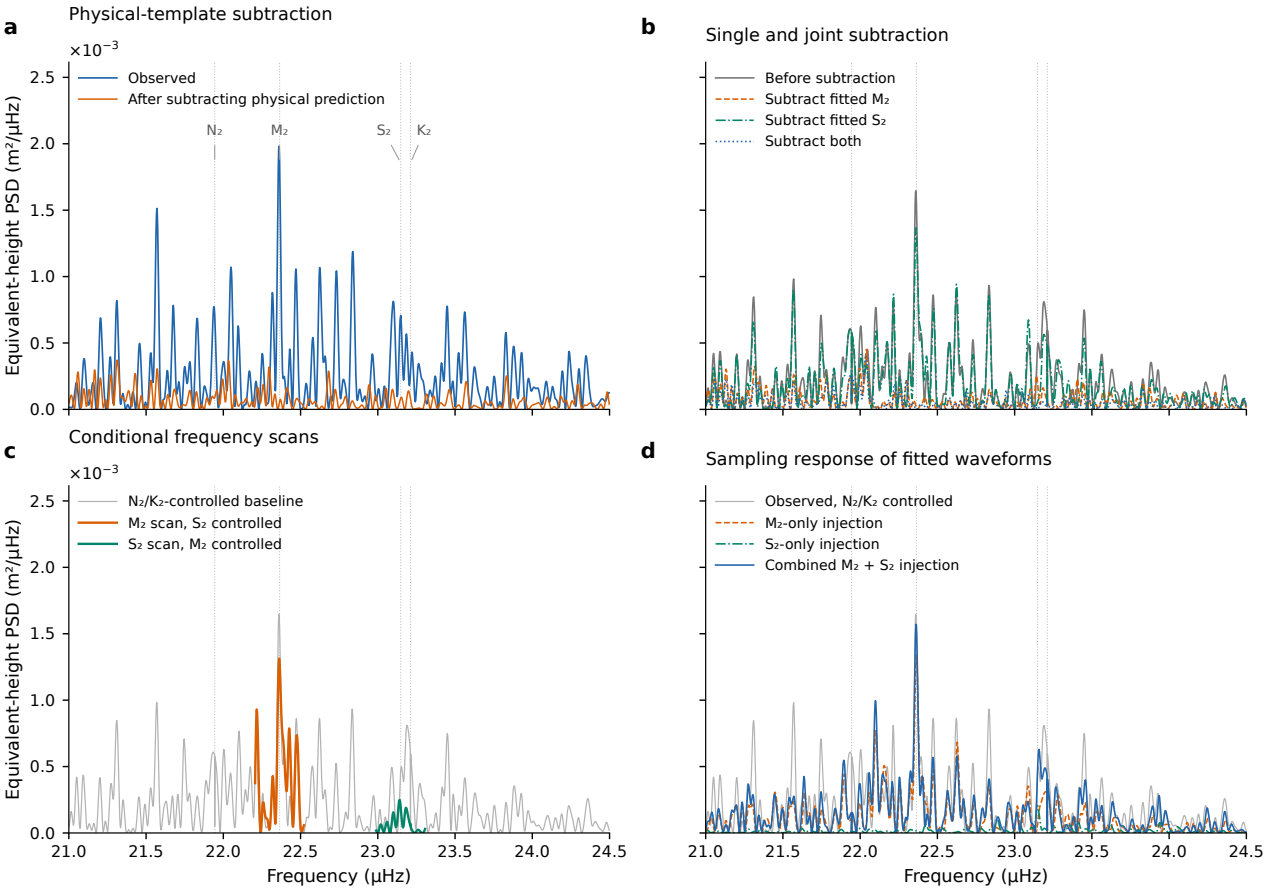
Extended Data Fig. 4 | April campaign in calendar time. Frequency-ratio observations after the 1,200-s triangular filter and removal of fitted segment offsets, their 3-h averages, and predictions with and without ocean loading. The horizontal axis retains actual calendar dates and gaps (local time, UTC+8).



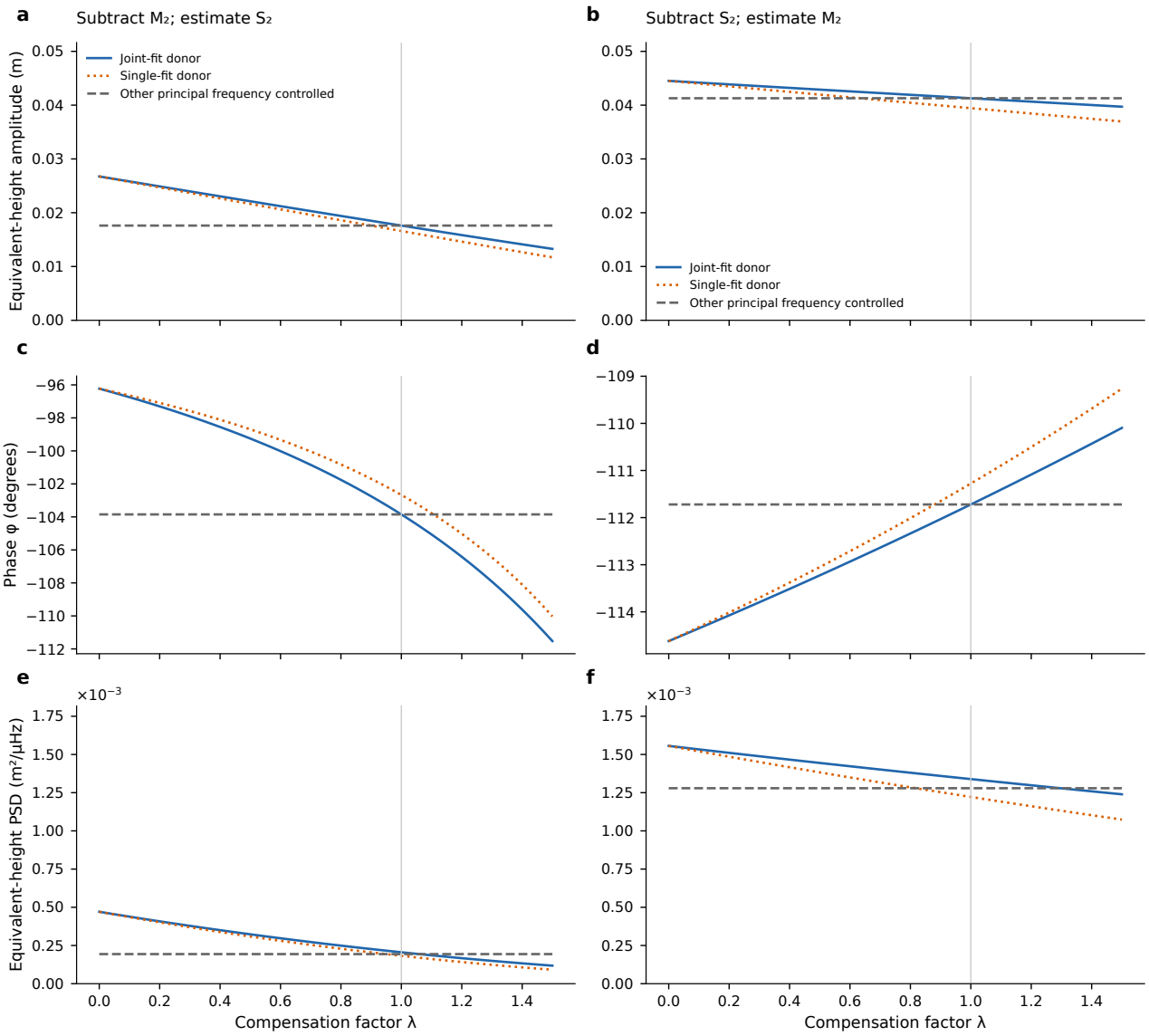
Extended Data Fig. 5 | Transient excursions and sample exclusion. **a,** Sixty-second averages around the excursion on 7 July 2026; red points indicate the 1-s samples excluded by the criterion in Methods. **b,** Distribution of deviations of 300-s means from a running median over ±30 min, divided by the robust standard deviation within each segment. The dashed line marks the threshold of six. The count axis is logarithmic. **c,** Observation time excluded in each campaign.



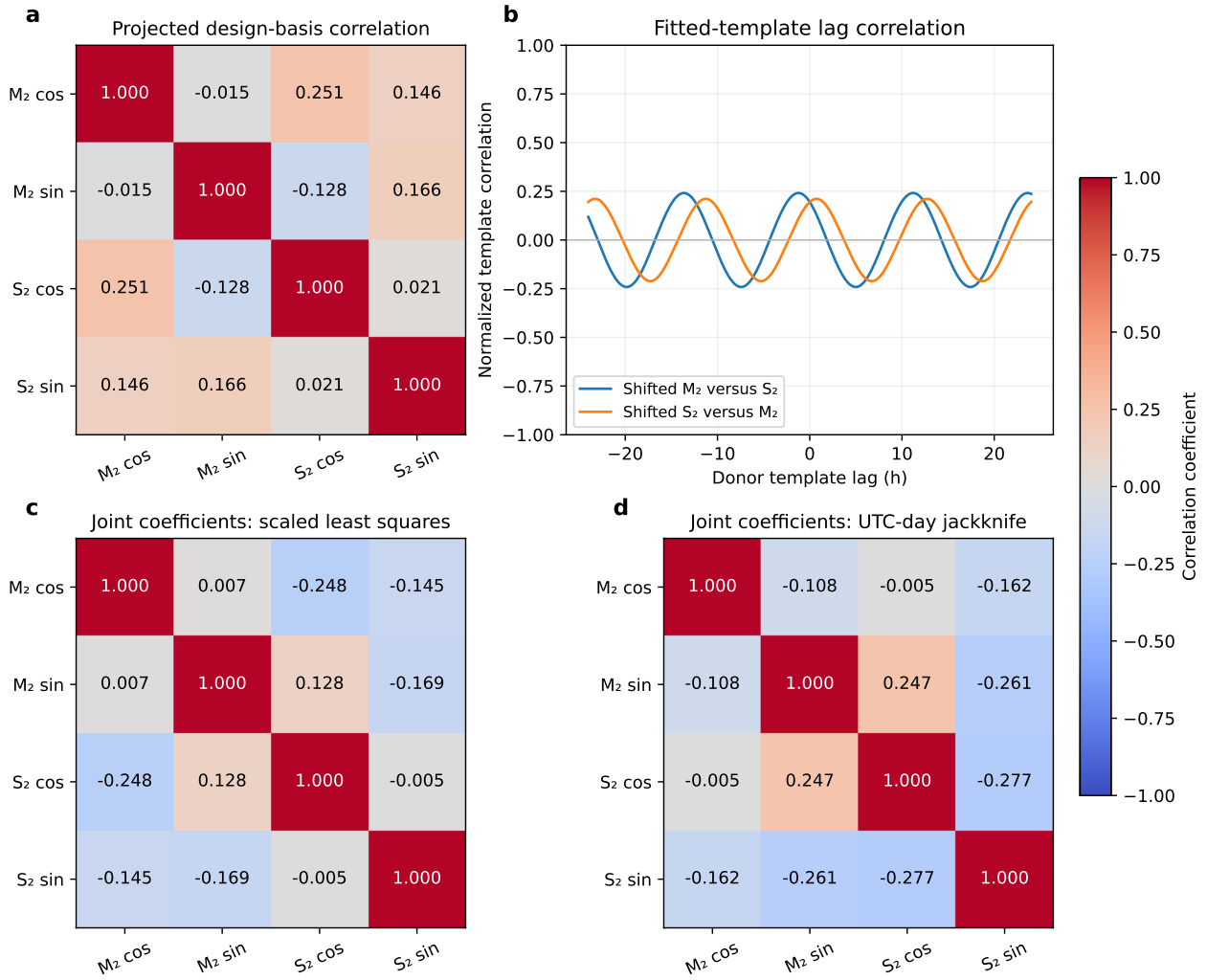
Extended Data Fig. 6 | Radial-displacement validation of the loading convolution. a,b, Amplitude ratios (**a**) and phase differences (**b**) between radial displacements calculated with the loading Green's function and the FES2022 load-tide grid in the centre-of-mass frame, at Shanghai and Wuhan, for eight constituents. Methods discusses the S_2 discrepancy.



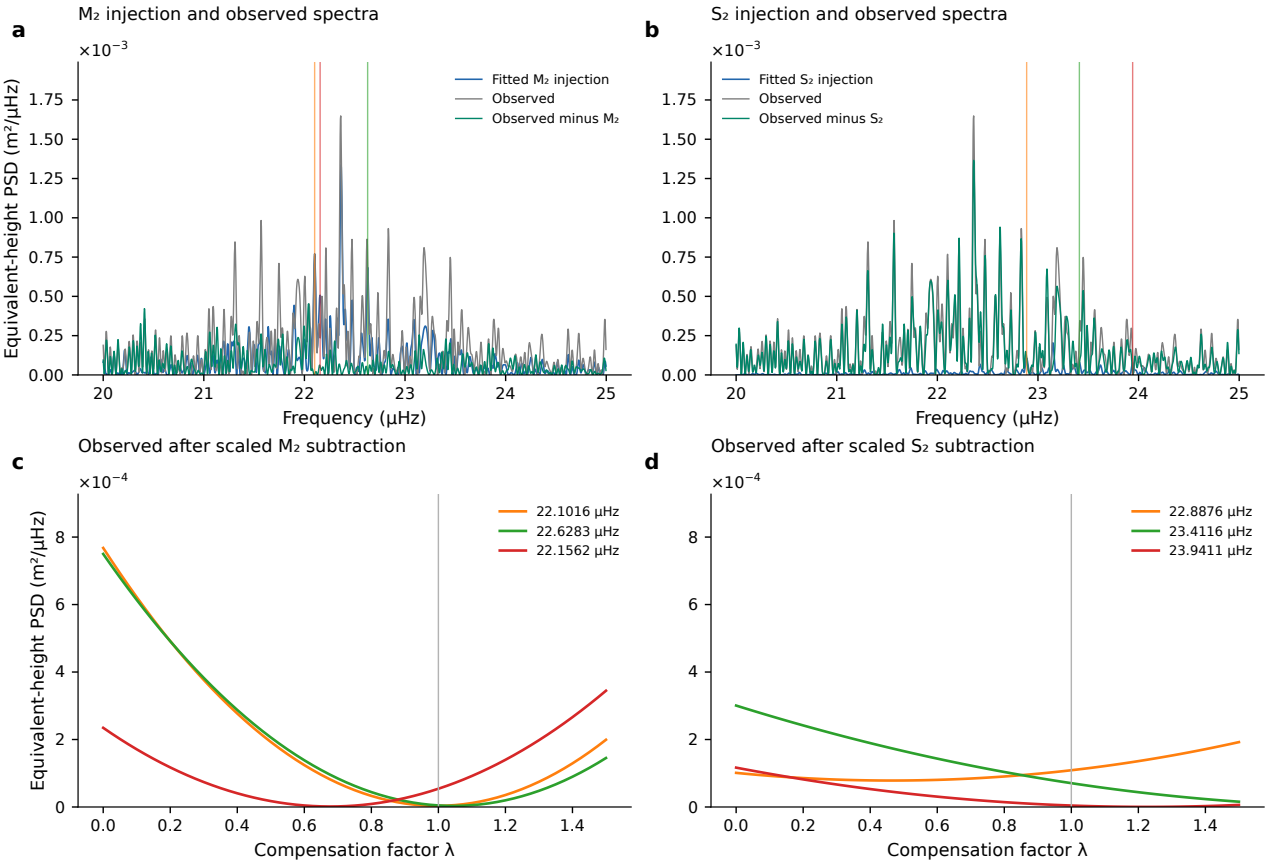
Extended Data Fig. 7 | Semidiurnal spectra and sampling response. All panels use the auxiliary non-overlapping 600-s means, their 3,299 recorded UTC bin-centre times, equal weights and 54 segment constants. **a,** Observed spectrum and spectrum after subtraction of the fixed unit-amplitude physical prediction, with segment constants controlled. **b,** Spectra before subtraction, after subtracting the fitted M_2 waveform, after subtracting the fitted S_2 waveform and after subtracting both. The waveforms come from a simultaneous freely phased M_2–S_2 fit controlling freely phased N_2 and K_2 terms. **c,** Scans near M_2 conditional on S_2, and near S_2 conditional on M_2, with N_2 and K_2 controlled in both cases. The grey curve is the N_2/K_2-controlled baseline. **d,** Spectra of deterministic M_2-only, S_2-only and combined fitted-waveform injections evaluated at the actual observation times and passed through the same nuisance projection, with the observed spectrum in grey. Interference means that the separate injected powers need not sum to the combined power. Dotted lines mark N_2, M_2, S_2 and K_2. The shared linear ordinate is equivalent-height power spectral density in m²/μHz, using the nominal-600-s normalization defined in Methods. No values are interpolated across gaps. The injection curves characterize the sampling response of the specified waveforms; they do not assign the fraction of an observed peak caused by leakage.



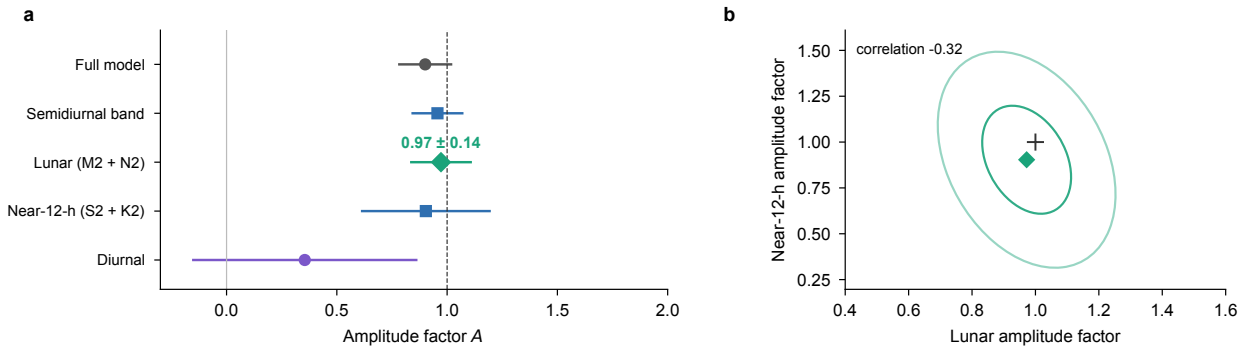
Extended Data Fig. 8 | Compensation sensitivity on the auxiliary 600-s record. a,c,e, Subtracting a scaled M_2 waveform and re-estimating S_2; b,d,f, the reverse. a,b, Target equivalent-height amplitude in metres; c,d, absolute phase in degrees; e,f, equivalent-height power spectral density in m²/μHz, using the nominal-600-s normalization in Methods. Solid curves use donor waveforms from the joint fit; dotted curves use single-frequency donor fits. All target-only fits retain segment offsets and free N_2/K_2 pairs. Dashed grey curves also control the donor pair and are invariant by construction. The power improvement for these curves is conditional on the donor and therefore has a different baseline from the target-only curves. $\lambda = 1$ denotes subtraction of the selected estimated waveform. Curves describe deterministic sensitivity conditional on the fitted waveforms. Amplitude and density panels share their respective vertical scales across directions, and density axes are linear with zero baselines.



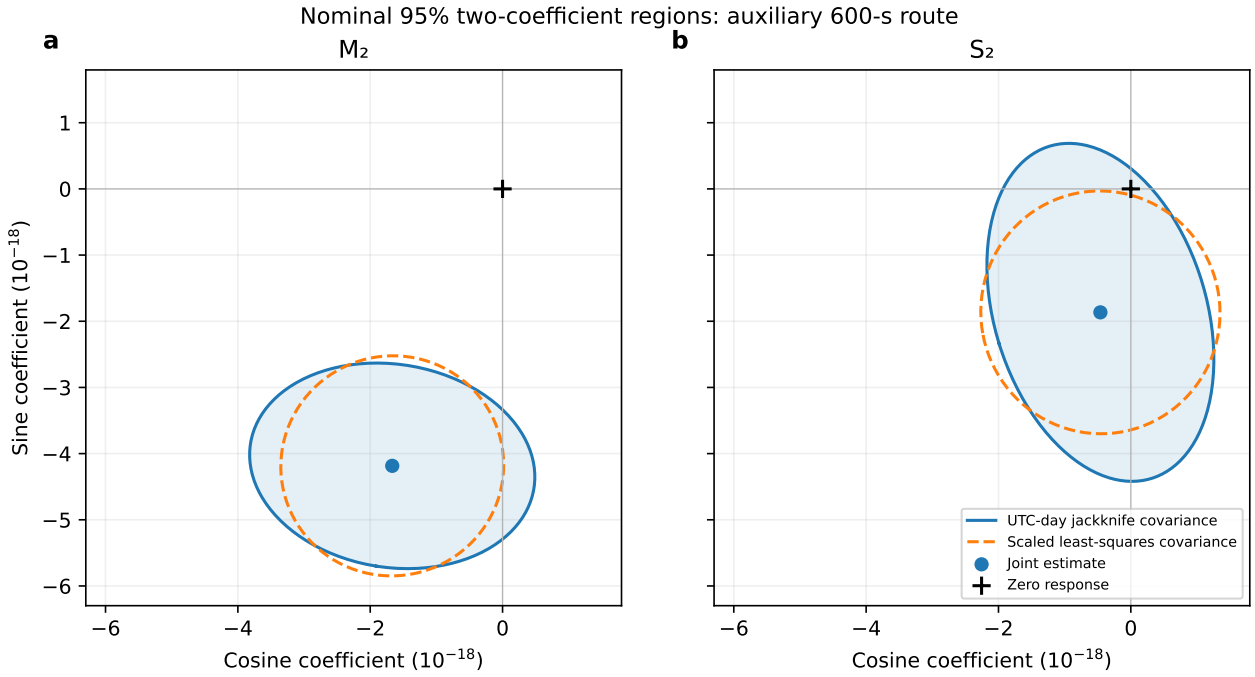
Extended Data Fig. 9 | Harmonic geometry and joint-coefficient correlations on the auxiliary 600-s record. a, Correlations of the M_2 and S_2 cosine and sine design columns after projecting out 54 segment constants and free N_2/K_2 pairs. b, Normalized lag correlations of the fitted templates, evaluated on the original timestamps without interpolating missing observations; these describe template similarity, not propagation delay. c,d, Correlations between the four coefficients estimated jointly, obtained by normalizing the existing scaled least-squares covariance (c) and UTC-day delete-one jackknife covariance (d). Coefficient order is M_2 cosine, M_2 sine, S_2 cosine and S_2 sine. The three matrices use the same −1 to +1 colour scale. Joint-coefficient covariance describes uncertainty coupling and is distinct from the design-column correlations in a. The 43 UTC days define deletion blocks for the jackknife, not 43 independent measurements. The joint correlations describe coupling between the estimated frequency components.



Extended Data Fig. 10 | Sampling aliases and compensation response. a,b, Observed spectra, spectra after subtraction of a joint-fit donor, and spectra of that donor alone on the identical timestamps and nuisance projection. Vertical markers identify three off-centre local maxima selected from the injection. c,d, Observed power at those fixed candidate frequencies as the compensation factor varies. The curves follow the quadratic responses predicted from the complex coefficient subtraction. Density is expressed in m²/μHz with the normalization in Methods. All density axes are linear and share scales within rows.



Extended Data Fig. 11 | Group responses and their covariance. a, Full, semidiurnal and jointly estimated group responses. Error bars are one standard deviation, calibrated by residual resampling for the full, semidiurnal and lunar estimates; neighbouring and diurnal errors use the larger of scaled least squares and the day jackknife. **b,** Covariance contours at radii one and two with the least-squares correlation scaled to the quoted marginal uncertainties.



Extended Data Fig. 12 | Joint coefficient uncertainty in the auxiliary 600-s analysis. a, M_2; b, S_2. Points are the free-phase coefficients from the joint M_2–S_2 fit with 54 segment offsets and free N_2/K_2 pairs. Solid blue ellipses and shading are nominal 95% two-coefficient regions obtained from the UTC-day jackknife covariance; dashed orange ellipses use scaled least-squares covariance. Both use radius $\sqrt{5.991}$ under a bivariate normal approximation. Black crosses mark zero response. The two panels have identical axes and equal aspect ratio. The jackknife M_2 region excludes zero, whereas its S_2 region includes zero. The regions use the existing coefficient covariances and the normal approximation described in Methods.

Extended Data Table 1 | Noise-model simulations of the alternative processing. Bias and coverage of nominal 95% intervals based on the largest of the least-squares, delete-day and delete-period standard errors. T denotes the full template without daily terms, and L the lunar group with daily terms, using the ten-day calendar-baseline design described in Methods and the actual observation times. Gaussian scenarios contain 1,024 simulations; Student-t and moving-average scenarios contain 4,096. OU denotes Ornstein–Uhlenbeck noise and MA(1) a first-order moving-average process.

Noise model	Bias (T)	Coverage (T), %	Bias (L)	Coverage (L), %
White	−0.000	96.7	−0.001	98.3
White + OU, 1 h	−0.007	94.5	−0.002	96.9
White + OU, 12 h	−0.002	96.1	+0.031	98.8
White + OU, 3 d	+0.003	95.9	+0.004	99.8
White + period constants	+0.001	96.0	+0.004	98.3
Student-t_7 MA(1)	+0.002	98.3	+0.004	98.2
Student-t_5 MA(1)	+0.001	98.9	+0.003	98.3
Student-t_7 MA(1) + OU	+0.001	97.2	+0.005	98.5

Extended Data Table 2 | Injection tests of the main estimator. Lunar amplitudes fitted with segment offsets and no empirical daily terms to resampled residuals plus either the full tidal model (Atrue = 1) or the model without the lunar group (Atrue = 0). The four daily-disturbance scenarios are defined in Methods, with 300 realizations per row. Bias and standard deviation describe the fitted amplitude. The median standard error is that computed within each realization, before the final residual-based calibration. Coverage is the fraction of intervals containing the injected amplitude; the accompanying 95% binomial interval refers to one-standard-error coverage. The last column gives the fraction with $A > 3\sigma$.

Disturbance	Atrue	Bias	s.d.	Median error	1σ coverage	95% binomial interval	1.96σ coverage	$A > 3\sigma$
None	0	−0.026	0.134	0.135	0.70	0.645–0.751	0.94	0.003
None	1	−0.003	0.135	0.137	0.71	0.659–0.764	0.94	1.000
24 h, air-temperature phase	0	−0.013	0.141	0.136	0.61	0.549–0.662	0.93	0.000
24 h, air-temperature phase	1	−0.007	0.145	0.138	0.64	0.579–0.691	0.94	1.000
Common 24 h + 12 h	0	−0.013	0.148	0.140	0.63	0.573–0.685	0.90	0.000
Common 24 h + 12 h	1	−0.024	0.148	0.139	0.60	0.546–0.659	0.92	1.000
Summer 24 h + 12 h	0	−0.021	0.138	0.140	0.67	0.610–0.720	0.95	0.000
Summer 24 h + 12 h	1	−0.016	0.150	0.137	0.59	0.529–0.643	0.92	1.000

Extended Data Table 3 | Daily terms and campaign-specific tidal amplitudes. Amplitude and local time of maximum of the 24-h term, and amplitude of the 12-h term, fitted jointly with the full tidal model within each campaign. Daily amplitudes are expressed in 10^{-18}. Their uncertainties are the larger of the scaled least-squares and day-by-day jackknife values; uncertainties in the time of maximum are jackknife estimates. Weak sinusoidal amplitudes are biased positive. The full-template amplitude A, fitted without daily terms, and its residual-calibrated uncertainty are those in Extended Data Fig. 3a. The last column gives the lunar amplitude fitted independently to each campaign with the main estimator.

Campaign	Days	24-h amplitude	24-h maximum, h	12-h amplitude	A	Alunar
Sep–Oct 2025	4	5.1 ± 2.7	20.7 ± 5.6	5.0 ± 9.7	1.28 ± 0.31	0.04 ± 0.74
Jan 2026	4	2.1 ± 3.0	17.2 ± 3.9	4.1 ± 1.8	1.27 ± 0.30	1.36 ± 0.37
Apr 2026	5	2.9 ± 2.2	1.9 ± 1.4	1.0 ± 3.9	0.91 ± 0.13	0.66 ± 0.42
May 2026	4	5.2 ± 3.3	3.8 ± 2.4	3.1 ± 2.6	0.97 ± 0.18	0.28 ± 0.68
Jun–Aug 2026	22	2.8 ± 1.3	15.8 ± 1.5	0.6 ± 0.7	0.67 ± 0.18	0.87 ± 0.18

Extended Data Table 4 | Free sinusoidal coefficients in the auxiliary reduction. Joint M_2 and S_2 cosine–sine coefficients, after controlling 54 segment constants and freely phased N_2 and K_2 terms. Coefficients and standard errors are in units of 10^{-18}. Phases refer to 6 September 2025, 16:05 UTC. Scaled least-squares errors use the residual variance; UTC-day jackknife errors delete each of the 43 observed dates and refit all coefficients. The full covariance matrices are retained with the analysis. These errors describe this auxiliary harmonic model and are not substituted for the primary lunar-group uncertainty.

Coefficient	Estimate	Scaled LS standard error	UTC-day jackknife standard error
M_2 cosine	-1.667	0.688	0.880
M_2 sine	-4.185	0.679	0.635
S_2 cosine	-0.460	0.738	0.701
S_2 sine	-1.866	0.750	1.043